Linux File Structure

dir/file	usage
`/etc/hosts`	local “DNS”
`/etc/passwd`	user config
`/etc/shadow`	password hashes
`/home`	user homes (for those that have homes)
`/tmp`	“temporary” files, usually located on RAM
`.`	current dir
`..`	parent dir
`~`	home dir of current user

Bash Basics

variable	usage
HOME	used for ~
USER	current username
PATH	paths for dirs with apps
SHELL	default shell
PWD	present working directory

use `$<variable>` to reference them

information command	usage
`man <command>`	system’s manual pager
`<command> --help`	internal manual
`history`	command history
`file <file>`	determine file type

navigation command	usage
`pwd`	present working directory
`cd <dir>`	change directory
`ls [<dir>]`	list content of dir
`ls -la [<dir>]`	... with hidden files and detailed info

management command	usage
`cp <src> <dst>`	copy src to dst
`mv <src> <dst>`	move src to dst
`rm <file>`	remove file
`rm -rf <file/dir>`	remove recursively with force (also dirs)
`cat <file>`	print file to standard output
`more <file>`	print with pager

search command	usage
`which <executable>`	locate path
`grep "regex" <file>`	g/re/p: global, reg. ex., print; search for pattern in file

search command	usage
`find <path> -name <filename>`	search recursively by name
`find <path> -type <filetype>`	search recursively by type
`find <path> -size <filesize>`	search recursively by size

pipe: `cat data.txt | grep "keyword"` - creates temporary storage to chain commands

asterisk: `ls *.txt` - matches any file or dir

escape special characters like: `echo "He said \"Hi\""` - use `"` or `'` to print spaces `echo "Hello world!"` - or `echo 'He said "Hi"'`

Stream Control

command	usage
`cmd > file`	STDOUT -> new file
`cmd >> file`	STDOUT -> append file
`cmd < file`	file -> STDIN

search for patterns: `<cmd> | grep "keywords"` sort lines: `<cmd> | sort` remove adjacent duplicates: `<cmd> | sort | uniq` keep only lines appearing once: `<cmd> | sort | uniq -u` map characters: `<cmd> | tr 'a-z' 'A-Z'`

Permissions

read, **write**, **execute** for **user**, **group**, **other**

command	usage
`whoami`	get username
`groups`	get groups of current user
`id`	get UID and GID's
`su [<user>]`	switch user (default root)
`sudo <cmd>`	execute cmd with root
`chown [<user>]:[<group>] <file>`	change ownership
`chmod 600 <file>`	change permissions (to `rw-------`)
`chmod u=rw,g=r,o=r <file>`	set permissions `rw-r--r--`

Process Management

command	usage
`ps`	processes associated with current shell
`ps aux`	snapshot of all current processes
`top`	`ps aux`, but updated
`sleep <n>`	sleep for n seconds
`<cmd> &`	run command in background
`jobs`	show background processes
`bg [<pid>]`	background current process or pid
`fg [<pid>]`	foreground latest process or pid
`kill <pid>`	send SIGTERM to pid
`kill -9 <pid>`	send SIGKILL to pid

systemctl for services/daemons

Networking Commands

command	usage
`ping [-c <n>] <host>`	ping host (n times)
`traceroute <host>`	show hops
`dig <host>`	dns query for host
`ip [-br] a`	list interfaces (short format)
`ss -tulpn`	show listening ports
`tcpdump -i <interface>`	capture traffic on interface

Transfer and Communication

get http via command line - `wget <url> -O <file>` - `curl <url> -o <file>`

unencrypted communication: `nc <url> <port>` ssh: `ssh [<user>@]<host> [-i <id_file>]`

Shortcuts

- for the default kali terminal: QTerminal

shortcut	usage
CTRL + SHIFT + C	copy selection
CTRL + SHIFT + V	paste clipboard
CTRL + C	SIGTERM to current process
CTRL + D	send EOF (close terminal, python, ...)
CTRL + Z	suspend current process
Up Arrow Key	cycle through previous commands in history
Down Arrow Key	cycle through next commands in history
Tab	auto-complete
Tab twice	list all possible completions

tmux

command	usage
`tmux [new -s <name>]`	start new session (named)
`tmux ls`	list sessions
`tmux a`	attach most recently detached session

modified config by Vocke in his blog

shortcut	vocke	usage
CTRL + B	CTRL + A	prefix (release before actual command)
C		create new window
W		display all open windows
{0,...9}		display window 0,...9
”	-	split horizontally
%	|	split vertically
X		close current pane
D		detach from current session
,		rename window
arrow	ALT + arrow	switch between panes

shortcut	vocke	usage
z		toggle zoom active pane